# OIT 277: Digital Platforms in the Age of AI

Stanford Graduate School of Business | Spring 2026

## Instructor

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## Course Assistants

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# Course Description

Digital platforms make millions of automated decisions every day: which driver to dispatch, which listing to show first, what price to charge, and which reviews to highlight and trust. This course examines how those decisions are made, why they matter for business performance, and how AI is transforming them. We cover the core design and operational pillars of digital platforms and examine how they shape user experience across applications such as matching and prediction markets. Finally, we discuss how AI agents may reshape markets and consider the implications for platform strategy, design, and governance.

### Hands-On Component

Students use AI tools (Claude, ChatGPT, Google Colab, Airtable) to prototype platform systems. Students write prompts and logic, not production code, so no prior coding experience is strictly necessary.

### Engagement Required

The labs demand genuine engagement. Students should expect to invest meaningful time in and outside of class both mastering the conceptual material and iterating on their prototypes.

### The Goal

The goal is not to produce polished demos but to develop the intuition for how these systems work, where they fail, and what tradeoffs they involve.

*This syllabus is tentative. Given the pace at which AI capabilities and platform strategies are evolving, specific topics, readings, and exercises may be adjusted during the quarter.*

# Course at a Glance

Nine sessions spanning lectures, hands-on labs, guest speakers, and team presentations.

| 1 | 2 | 3 |
|---|---|---|
| **Introduction**<br>**April 2**<br>Lecture + Exercise | **Matching Under Uncertainty**<br>**April 6**<br>Lecture + Hands-On Lab | **Guest Speaker**<br>**April 9**<br>W Zhang (LinkedIn, VP) |
| **4** | **5** | **6** |
| **Search and Personalization**<br>**April 13**<br>Lecture + Hands-On Lab | **Pricing and Monetization**<br>**April 16**<br>Lecture + Hands-On Lab | **Prediction Markets**<br>**April 20**<br>Lecture + Hands-On Lab |
| **7** | **8** | **9** |
| **Guest Speaker**<br>**April 23**<br>TBA | **AI Agents**<br>**April 27**<br>Lecture + Build Session | **Team Projects**<br>**April 30**<br>Presentations & Discussion |

# Evaluation & Grading

## Grade Breakdown

**40%**

**Team Project**

Proposal + prototype + presentation

**30%**

**Lab Assignments / Homework**

**30%**

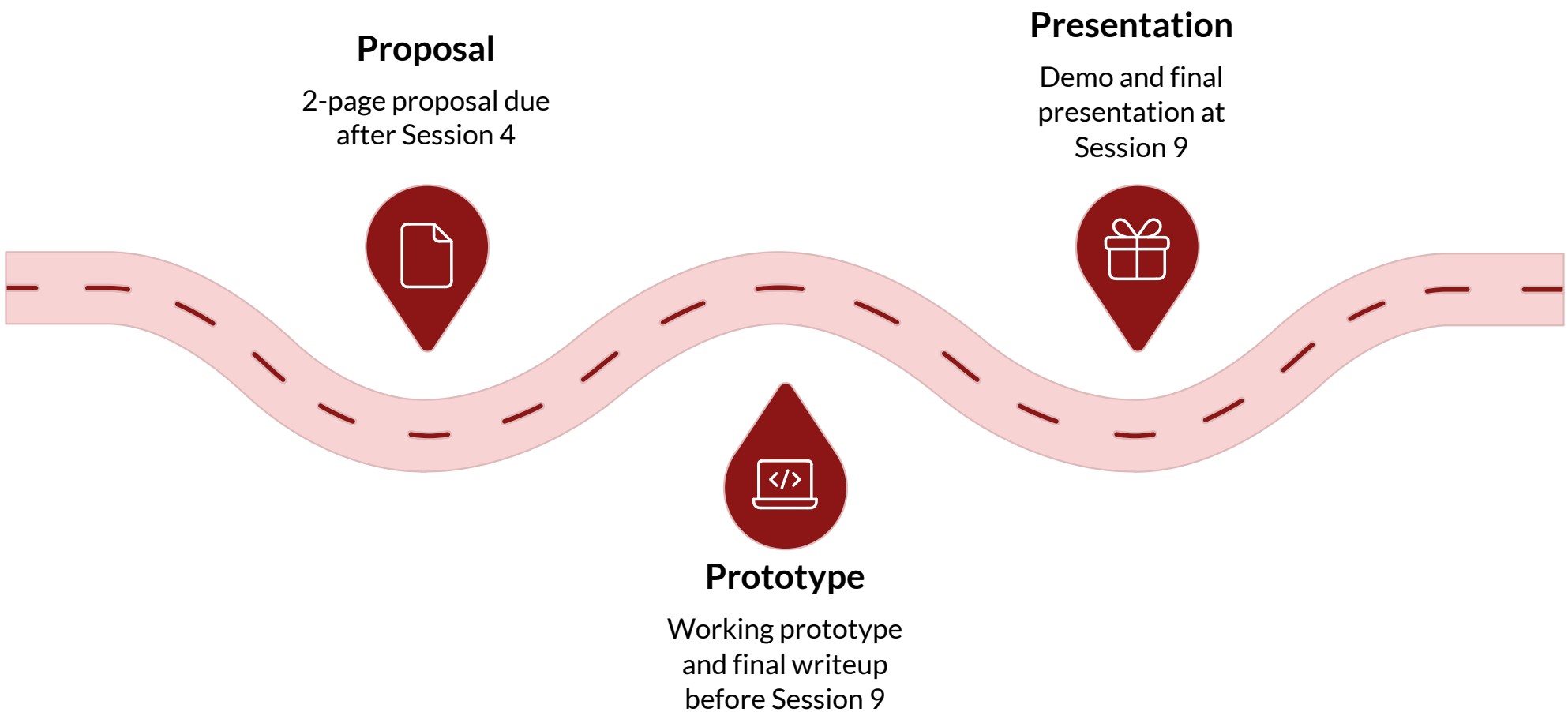
**Class Participation**

## Team Project Requirements

Completed projects should include:

- A working prototype component of a platform/marketplace
- Key design decisions and the tradeoffs they involve
- Where the AI works well and where it fails

# Project Timeline




Each milestone builds toward a complete platform prototype. The writeup should cover the problem, design, prototype, evaluation, limitations, and a validation section. The final presentation should demo the prototype with a focus on decisions and tradeoffs, not polish.

# Attendance & Participation Policies

## Attendance

Attendance is recorded at the start of each session. Late arrivals count as absences. Each student is allowed one absence without penalty; please notify the instructor and CA by email in advance. Beyond that first absence, each additional missed session will reduce your grade.

## Class Participation

The participation grade rewards substantive contribution to class discussion. **Quality matters more than quantity.** Guest speaker sessions carry the same participation expectations as regular lectures.

| 2 Absences | 3 Absences | 4+ Absences |
|---|---|---|
| No Participation Grade | Low Pass | Result in a Fail |

# Course Policies: Submissions & Teams

### Late Submissions

Assignments and project deliverables are due by the deadlines posted on Canvas. Submissions more than 1 day late will not be accepted. If you face an extenuating circumstance, contact the instructor before the deadline.

### Team Formation

Project teams consist of 4 students from the same section. Teams should be formed and reported to the CA by the end of Session 2. If you are unable to find a team, contact the CA and we will help match you.

# Use of AI Tools & Honor Code

## Use of AI Tools

AI assistants (Claude, ChatGPT, etc) are permitted and encouraged for all coursework in this class. The labs are designed around them. However, **you must be able to explain and defend any work you submit.** You should be able to walk through your reasoning, describe what the AI contributed, and identify where its output was wrong or incomplete.

## Honor Code

All submissions are subject to the Stanford Honor Code. You may collaborate within your project team on team deliverables. Individual lab and homework assignments are individual work. If you are uncertain whether a particular form of collaboration is permitted, ask before proceeding.

*Submitting AI-generated work you do not understand is a violation of the spirit of this course*

# Classroom Norms

### Session Continuity

Sessions run without a break. If you need to leave early, let the instructor know in advance; leaving without notice counts as an absence.

### Name Cards

Please use your name card.

### Laptops & Tablets

Laptops and tablets are welcome during lab sessions and for note-taking, but not for unrelated browsing or email.

### Phone Use

Phone use during class will affect your participation grade.

# Tools

Setup guides will be posted on Canvas before the first lab session.

### Claude / ChatGPT

AI tool for building and stress-testing prototypes in labs and the final project.

### Google Colab

Environment for running AI-generated scripts.

### Airtable

Structured data management and workflow prototyping.